8 Fig. 8.1 shows a metal rod suspended in the magnetic field produced by a pair of permanent magnets. The metal rod is connected to a cell and there is a current in the metal rod.

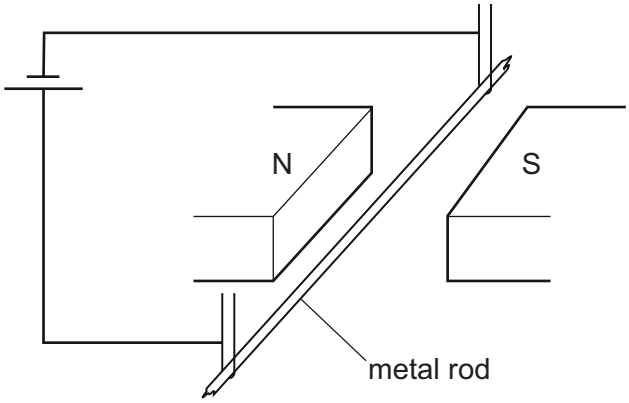

Fig. 8.1

(a) State the direction of the force on the metal rod due to the current.

Explain your answer.

direction of force .....

explanation .....

.....

.....

[3]

(b) The connections to the cell are reversed.

State how this change affects the force on the metal rod.

..... [1]

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(c) Two magnets and a cell are used to make a simple electric motor as shown in Fig. 8.2.

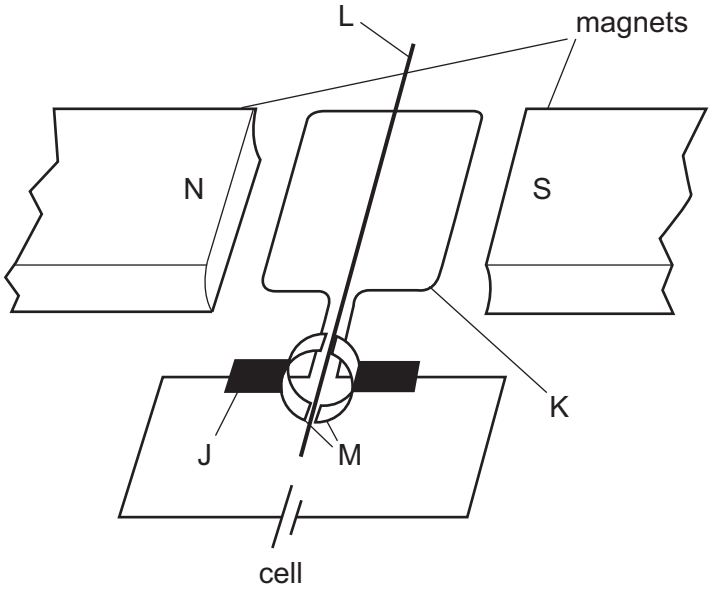

Fig. 8.2

Describe the function of parts J, K, L and M.

J .....

.....

K .....

.....

L .....

.....

M .....

.....

[4]

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